

# Index of Research Works

Titles, abstracts, and keywords

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## Gauge-Persistence Transform for Yang-Mills Gradient Flow: A flow-indexed topological invariant on gauge quotients

**Abstract.** We define a gauge-invariant persistence construction for Yang–Mills configurations by sampling the Yang–Mills gradient flow orbit of a gauge class, forming a finite metric point cloud in the quotient space, and computing its Vietoris-Rips persistence diagrams. The resulting object is a flow-indexed barcode-valued transform. We define derived observables such as total persistence, persistence entropy, and vacuum isolation radius.

**Keywords:** Yang-Mills theory, gradient flow, persistent homology, gauge quotients, topological data analysis

## Scale-Invariant Structural Compactness and Chazy-Singularity Attractors in Non-Stationary Protein Folding Dynamics

**Abstract.** Traditional predictive models of protein conformational transitions rely on high-dimensional Molecular Dynamics (MD), scaling at  $O(2^N)$ . We present a reduced-order analytical framework that decouples topological compactness from physical size using a Gamma-Tilted Completely Monotone Mixture (GTCM). We map sequential amino acid electrostatic interactions to continuous fluid-dynamic potential fields, reducing folding pathways to a 3rd-order non-linear chaotic attractor (Chazy equation). This yields an  $O(1)$  diagnostic threshold for protein decoherence and misfolding, providing deterministic transition times ( $t_{unfold}$ ) relevant for neurodegenerative pathology and synthetic biology.

**Keywords:** protein folding, reduced-order modeling, Chazy equation, chaotic attractors, misfolding diagnostics

## Reduced-Order and Spectral-Geometric Diagnostics of Protein Conformational Transitions

**Abstract.** We study protein conformational change through a unified reduced-order framework for folding, misfolding, and structural failure. The model replaces exhaustive molecular sampling with a compact nonlinear evolution law that detects transition thresholds directly in state space, separating stable, metastable, and unstable regimes. In parallel, we characterize the same conformational transitions through spectral and geometric features of high-frequency molecular signals, enabling rapid biosensing of decoherence and loss of structural integrity.

**Keywords:** protein conformations, reduced-order dynamics, spectral geometry, biosensing, stability thresholds

## Spectral Zeta Moments and Carleman Indeterminacy in Four-Dimensional Gauge Theory

**Abstract.** We identify a fundamental obstruction to a class of spectral approaches to the Yang–Mills mass gap by recasting the problem as a Stieltjes moment problem for logarithmic spectral

data. In four spacetime dimensions, the associated moment growth is incompatible with Carleman determinacy, so the moment sequence does not uniquely determine the underlying spectral measure. As a consequence, distinct spectral configurations may share the same zeta-derived moment data, including configurations with and without a mass gap. This shows that moment-theoretic information extracted from spectral zeta derivatives is insufficient, by itself, to resolve the mass-gap question in four-dimensional gauge theory.

**Keywords:** spectral zeta functions, Carleman indeterminacy, moment problems, Yang-Mills theory, mass gap

## Quasiperiodic Structure on the Critical Line of Spectral Zeta Functions: Log-Spectral Encoding of Eigenvalue Gaps

**Abstract.** We study the spectral zeta function on the critical line as a quasiperiodic signal generated by the logarithmic distribution of eigenvalues. In this picture, the spectral data admit a frequency-domain interpretation in which the presence of a mass gap is reflected by a dominant low-lying mode separated from the remainder of the spectrum. The resulting formulation yields an explicit lower bound for the critical-line amplitude in the regime where the first eigenvalue dominates the spectral sum. This recasts gap detection as a problem in harmonic analysis on logarithmic spectral data, and highlights a new bridge between operator theory, signal structure, and spectral geometry.

**Keywords:** spectral zeta, critical line, quasiperiodicity, eigenvalue gaps, harmonic analysis

## Topological Persistence and Spectral Gaps in Yang-Mills Theory: A Reformulation of the Mass Gap

**Abstract.** We present a topological reformulation of the Yang-Mills mass gap problem in which the existence of a positive spectral gap is encoded by persistence of the lowest-energy mode under the spectral filtration. In this formulation, the gap condition is equivalent to a family of stability statements connecting the spectral, topological, and analytic structure of the theory. The result identifies a geometric criterion for mass-gap detection that places the problem in a broader framework of persistent homology and analytic continuation. This recasts the Yang-Mills mass gap not merely as an operator-theoretic question, but as a structural constraint on how the vacuum sector survives under spectral deformation.

**Keywords:** Yang-Mills, mass gap, spectral filtration, persistent homology, vacuum stability

## Borel Resummation and Complex-Analytic Stability Bounds in Non-Perturbative Quantum Field Theory

**Abstract.** We study the analytic continuation of divergent perturbative series through the lens of Borel regularization and complex stability bounds. The leading contribution defines a nonvanishing analytic envelope, while the remaining correction term determines whether the continuation remains stable on the relevant contour. This yields a sharp criterion separating the controlled regime from the unresolved regime and identifies a critical coupling beyond which the nonperturbative question

remains open. The result situates the mass gap problem within a broader analytic framework connecting divergent series, contour stability, and regularized summation phenomena.

**Keywords:** Borel resummation, non-perturbative QFT, analytic continuation, stability bounds, divergent series

## Representation Gaps in Major Open Problems: Coordinate Charts, Flat Projections, and the Loss of Nonlinear Structure

**Abstract.** We isolate a common structural issue in leading approaches to several major open problems: the use of representations that are locally consistent but globally incomplete. In the cases of Yang–Mills theory and additive prime problems, standard perturbative or asymptotic frameworks capture only a flattened image of a deeply nonlinear object, thereby losing the features most relevant to the unresolved question. We describe this failure as a representation gap, and argue that progress requires coordinate systems or atlases that preserve the relevant nonlinear and global structure. The result is a unifying conceptual diagnosis of why familiar analytic tools may fail at exactly the point where the problem becomes difficult.

**Keywords:** representation gap, nonlinear structure, coordinate charts, Yang-Mills theory, prime problems

## Finite-Dimensional Autonomous ODE Representations of Laplace-Transformed Arithmetic Functions

**Abstract.** We characterize arithmetic functions whose Laplace-type generating functions satisfy finite-dimensional autonomous ordinary differential equations. For a broad class of multiplicative arithmetic signals, the induced dynamics close on finitely many scalar modes, with the dimension determined by divisor structure. This yields a classification that unifies several classical examples and shows that arithmetic weights can be encoded by low-dimensional nonlinear dynamics. The result opens a systematic connection between analytic number theory, generating functions, and dynamical systems.

**Keywords:** arithmetic functions, Laplace transform, autonomous ODEs, multiplicative signals, divisor structure

## Spectral Obstruction to Hilbert-Pólya Realizations via Positive Laplace Kernels

**Abstract.** We identify a structural obstruction to a natural class of operator-based approaches to the Riemann Hypothesis. For positive Laplace-type kernels with the required zeta-linked transform structure, the spectral weights are too rigid to encode the multiplicative information needed to distinguish prime structure from ordinary integer growth. This yields a no-go result for a broad family of Hilbert-Pólya realizations and clarifies where any successful construction must depart from the additive spectral framework.

**Keywords:** Hilbert-Pólya, Riemann Hypothesis, Laplace kernels, spectral obstruction, positive operators

## The Instanton Trans-Series as a Generational Decomposition: A Structural Isomorphism in Resurgence Theory

**Abstract.** Resurgence theory expresses the Yang–Mills partition function as a trans-series indexed by instanton number  $k$ . We identify a structural isomorphism between this instanton decomposition and a generational prime decomposition framework in additive number theory, in which the instanton sector of charge  $k$  corresponds to the  $k$ -th prime generation.

**Keywords:** resurgence theory, instantons, trans-series, Yang–Mills partition function, prime generations

## Rouché Envelope Bounds and TDA Persistence in the Yang–Mills Mass Gap

**Abstract.** We present a unified structural formulation of the Yang–Mills mass gap problem in which spectral stability, topological persistence, and analytic envelope bounds are treated as equivalent manifestations of the same underlying condition. The result yields a geometric criterion for the presence of a gap, while also identifying a moment-growth obstruction to purely spectral reconstruction methods in four dimensions. This places the mass gap problem within a broader framework of nonlinear stability, spectral analysis, and topological diagnostics.

**Keywords:** Rouché theorem, envelope bounds, persistent homology, Yang–Mills mass gap, spectral stability

## Arithmetic Reduction of Network Contagion Dynamics on Scale-Free Topologies

**Abstract.** We develop an arithmetic model for contagion on structured scale-free networks in which macroscopic propagation is governed by a finite set of scalar modes. This reduction yields an explicit criterion for the onset of rapid spread and replaces large-scale graph-based computation with a compact dynamical description. The exact reduction applies to networks with arithmetic transmission structure, while broader cases admit controlled approximation. The work suggests a new computational route for structured diffusion processes in complex networks.

**Keywords:** network contagion, scale-free networks, arithmetic reduction, diffusion dynamics, complex networks

## Arithmetic Matched Filtering for Structure-Based Detection of Non-Stationary Signals

**Abstract.** We introduce a signal-detection framework for non-stationary environments in which the relevant structure is identified directly from arithmetic mode content rather than carrier frequency. The method supports efficient streaming detection, remains effective under varying phase and modulation, and provides a robust alternative to standard Fourier-based approaches when temporal regularity is present but frequency content is unstable. The analysis also gives a principled false-alarm framework under noise and motivates real-time implementations for structured signals.

**Keywords:** matched filtering, non-stationary signals, arithmetic structure, streaming detection, false alarms

## Finite-Key Security Bounds for an Asymmetric Adversary Model in Device-Independent or Prepare-and-Measure QKD

**Abstract.** We prove a sharp finite-key theorem for secret-key extraction from asymmetric correlated sources, where the legitimate parties observe unequal noise and the adversary has bounded side information. The result gives an explicit lower bound on the extractable key length together with a composable secrecy guarantee and a correctness guarantee. It explains when short, imperfect, and asymmetric data can still be converted into uniform secret key material, and it has direct applications to quantum key distribution, threshold secret sharing, leakage-resilient key management, and distributed authentication.

**Keywords:** quantum key distribution, finite-key analysis, composable security, asymmetric noise, secret-key extraction

## Quantum Sensing: Unruh-Bogoliubov Thermal Filtering

**Abstract.** *Abstract not supplied in the source list.*

**Keywords:** quantum sensing, Unruh effect, Bogoliubov transformations, thermal filtering, relativistic sensing

## Quantum Sensing: Rindler-Bogoliubov Equivalence Proxy

**Abstract.** *Abstract not supplied in the source list.*

**Keywords:** quantum sensing, Rindler coordinates, Bogoliubov transformations, equivalence principle, proxy model

## Quantum Sensing: Vacuum State Transitions

**Abstract.** *Abstract not supplied in the source list.*

**Keywords:** quantum sensing, vacuum states, state transitions, field fluctuations, sensing

## Quantum Sensing: Atomic Clock Decoherence / IMU Noise

**Abstract.** *Abstract not supplied in the source list.*

**Keywords:** quantum sensing, atomic clocks, decoherence, inertial measurement units, noise modeling

## Signal/Filtering: $O(N)$ Non-Markovian Noise and Adaptive Thresholding

**Abstract.** *Abstract not supplied in the source list.*

**Keywords:** signal processing, non-Markovian noise, adaptive thresholding,  $O(N)$  algorithms, filtering

## Log-Periodic Oscillations and Spectral Dimension Locking in Self-Similar Laplacian Systems

**Abstract.** Self-similar Laplacian operators on fractal domains exhibit heat-trace asymptotics that deviate from the classical Weyl law by a multiplicative log-periodic correction. We prove that this correction locks the spectral dimension to a rational value determined entirely by the self-similarity data.

**Keywords:** fractal Laplacians, log-periodic oscillations, spectral dimension, heat trace, self-similarity

## A Susceptibility Curvature Law for Perturbed Landau-Level Systems

**Abstract.** Landau quantization produces an equally spaced energy ladder

$$E_n = \hbar\omega_c(n + 1/2)$$

. We prove that small perturbations – disorder, confinement, or strain – generate a universal curvature correction to magnetic susceptibility.

**Keywords:** Landau levels, magnetic susceptibility, perturbation theory, curvature law, quantum magnetism

## A Complete Anharmonic Correction Hierarchy for Near-Harmonic Molecular Vibrational Spectra

**Abstract.** Near-harmonic molecular potentials produce vibrational spectra that deviate from the harmonic ladder

$$E_n = \hbar\omega(n + 1/2)$$

by a hierarchy of corrections. We derive this hierarchy to all orders in perturbation theory using the Birkhoff normal form.

**Keywords:** molecular vibrations, anharmonicity, Birkhoff normal form, perturbation theory, spectral corrections



## Topological Unknotting of Magnetic Helicity for High-Altitude EMP Shielding

**Abstract.** Protecting critical infrastructure against high-altitude Electromagnetic Pulses (HEMP) typically relies on brute-force Faraday topologies, which are insufficient against intense, localized E1-phase plasma transients. We propose a Magnetohydrodynamic (MHD) shielding paradigm based on the topological conservation of Magnetic Helicity.

**Keywords:** EMP shielding, magnetic helicity, magnetohydrodynamics, topological methods, critical infrastructure

## Non-Stationary Drift Isolation in High-Dimensional Financial Diagnostics: A Thermodynamic Orthogonalization Approach

**Abstract.** Traditional Value-at-Risk (VaR) and alpha-generation models suffer from rapid decay when exposed to non-stationary market regimes, primarily due to the  $O(N^3)$  computational cost of dynamically inverting massive covariance matrices. We present a novel diagnostic framework mapping financial time-series to non-equilibrium thermodynamic drift. By tracking phase-space divergence via the Fokker-Planck equation, we orthogonalize alpha decay, providing an  $O(1)$  early-warning metric for quantitative portfolio degradation.

**Keywords:** financial diagnostics, non-stationarity, thermodynamic drift, Fokker-Planck equation, portfolio risk

## Quasi-Discrete Spectral Structure and Volatility Cascade Regimes in High-Frequency Market Microstructure

**Abstract.** We show that Hawkes process models of high-frequency order flow produce power spectral densities with quasi-discrete resonance structure when the excitation kernel has near-harmonic characteristic frequencies. This quasi-discreteness gives rise to volatility clustering as a spectral resonance cascade.

**Keywords:** market microstructure, Hawkes processes, volatility clustering, spectral resonance, high-frequency finance

## Cross-Frequency Phase-Locking Invariants in Coupled Neural Oscillator Networks

**Abstract.** We prove the existence of topological locking invariants for coupled neural oscillators whose natural frequencies form a near-harmonic hierarchy.

**Keywords:** neural oscillators, phase locking, topological invariants, coupled networks, frequency hierarchy

## SC-DKVL: Cryptographic Content-Addressed Distributed Mutual Exclusion

**Abstract.** A distributed locking mechanism that solves version-mismatch race conditions during rolling deployments. Instead of arbitrary resource strings, it uses precomputed Git blob hashes as lock identifiers, guaranteeing that differing code versions do not collide on critical sections. Statistical / Mathematical Method: Cryptographic Hashing (SHA-1), Distributed Consensus (Lua scripting in Redis). Output Metrics: Achieves 0.5 – 2 ms lock acquisition latency in same-datacenter deployments (Redis), outperforming standard etcd lease consensus (3 – 15ms) while adding strict code-version invariants.

**Keywords:** distributed locking, mutual exclusion, Git hashes, Redis, deployment safety

## Resilient AI-Powered Navigation via Domain-Driven Geospatial Persistence

**Abstract.** An architectural system design separating high-frequency Geospatial Tracking (C++ daemon) from transient Dialogue Management (Cloud LLM) using zero-copy shared memory. Prevents total location loss during API/network crashes. Output Metrics: Preserves continuous location tracking through 15-second network outages. Resynchronizes with the LLM client in <200 ms upon reconnection (compared to 5 – 8 second cold-start baseline delays) – a  $34\times$  faster recovery rate.

**Keywords:** AI navigation, geospatial persistence, zero-copy shared memory, resilience, system architecture

## 0/1 Knapsack: Dynamic Programming vs. Spectral Analytical

**Abstract.** A comparative implementation of the classical 0/1 Knapsack problem using standard Dynamic Programming (DP) versus a novel Spectral Analytical approach. The spectral method maps combinatorial states to algebraic exponents via generating functions (isomorphic to Casimir Zero-Point Energy Regularization), converting discrete choices into a continuous effective potential minimum. Statistical / Mathematical Method: Partition/Generating Functions, Polynomial Expansion, Zero-Point Energy Regularization (Analytical Mapping). Code Pseudo: `Z = 1; for w_i, v_i in zip(weights, values): Z = expand(Z * (1 + (y^w_i) * (z^v_i)))`

**Keywords:** 0/1 knapsack, dynamic programming, generating functions, spectral methods, combinatorial optimization

## A Geometric Framework for the Representation of Heterogeneous Truth States on Nested Spherical Manifolds

**Abstract.** We introduce a geometric framework for representing heterogeneous variables – continuous, discrete, and qualitative – within a unified state-space that does not rely on Euclidean coordinates or commensurate measurement axes. By treating truth as a localized selection within an outcome space and encoding outcome spaces as spherical manifolds whose radii scale with cardinality,

we derive a nested geometric structure in which probabilistic, informational, and metric properties arise as consequences rather than assumptions.

**Keywords:** geometric representation, heterogeneous states, spherical manifolds, truth states, unified state-space

## Substring Matching: KMP vs. Parametric Resonance

**Abstract.** Replacing the symbolic sequential branching of classical substring search (Knuth-Morris-Pratt) with a physics-first continuous wave-field model. The text is treated as a 1D periodic potential lattice and the pattern as a driving frequency, utilizing Floquet Theory and Anderson Localization to find exact matches via resonance spikes. Statistical / Mathematical Method: L2 Norm Minimization, Fast Fourier Transform (FFT) Cross-Correlation, Anderson Localization (Wave Function Collapse). Code Pseudo: `T_sq_sum = prefix_sum(T^2); P_sq_sum = sum(P^2); cross_term = fftconvolve(T, P_rev); H_int = T_sq_sum + P_sq_sum - 2 * cross_term; match_index = argmin(H_int)`

**Keywords:** string matching, KMP, parametric resonance, FFT cross-correlation, Anderson localization

## Graph Cycle Detection: DFS vs. Topological Trace Invariant

**Abstract.** Mapping directed graph cycle detection to a discrete gauge field connection. Instead of iteratively walking nodes via Depth-First Search, the algorithm evaluates the Chern-Simons global trace over the adjacency matrix to detect non-trivial holonomies (Wilson Loops), proving the existence of a cycle through a single macroscopic invariant. Statistical / Mathematical Method: Matrix Eigenvalue Decomposition, Chern-Simons Theory, Macroscopic Holonomy Trace. Code Pseudo: `Eigenvalues = ComputeEigenvalues(AdjacencyMatrix); HolonomyTrace = Sum(Abs(Eigenvalues)); is_cycle = HolonomyTrace > 1e-7`

**Keywords:** cycle detection, DFS, topological invariant, Chern-Simons theory, graph algorithms

## Spectral Graph Benchmark Suite

**Abstract.** A multi-domain evaluation of the Spectral Trace framework against classical industry baselines across five core computational problems: Feedback Intensity, Network Anomaly, Diffusion/Spread, Graph Similarity, and System Stability. Statistical / Mathematical Method: Spectral Trace, Structural EWMA/z-score, Krylov Subspace (expm\_multiply), 1-Weisfeiler-Lehman (1-WL) Refinement.

**Keywords:** benchmarking, spectral graph methods, anomaly detection, diffusion, graph similarity